E0X PBR Energy Balance Example

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Why Are We Learning This?

Many industrial reactors are **nonisothermal**, meaning that the reaction changes the temperature of the reacting mixture as it proceeds. In these systems, reaction rates and temperature influence one another continuously. This example introduces several important reactor-design concepts that appear throughout chemical reaction engineering:

- **Coupling of mole and energy balances**. Temperature affects the reaction rate, the reaction rate affects heat generation, and heat generation affects temperature.
- **Temperature-dependent kinetics**. The Arrhenius equation shows that even modest temperature changes can dramatically alter reaction rates.
- **Temperature-dependent equilibrium**. For reversible reactions, temperature affects not only how fast the reaction proceeds, but also the maximum conversion that can be achieved.
- **Heat-transfer effects**. Industrial reactors often use heat exchangers to improve performance, maintain safe operating conditions, and prevent reactors from approaching unfavorable equilibrium limitations.
- **Pressure-drop effects in packed-bed reactors**. Changes in pressure affect concentrations and, therefore, reaction rates and reactor performance.

By the end of this example, you should be able to build and solve a mathematical model that simultaneously predicts: Conversion, Equilibrium Conversion, Temperature, Coolant Temperature, and Reaction Rate as functions of catalyst weight. More importantly, you should understand **why different heat-exchange strategies lead to different reactor performances** and how reactor design balances kinetics, equilibrium, heat transfer, and pressure drop.

Key Engineering Insight

Reaction Rate
↓
Heat Generation
↓
Temperature Changes
↓
Rate Constant Changes
↓
Reaction Rate Changes
In a nonisothermal reactor, kinetics and energy balances form a feedback loop. The concentration, temperature, and conversion profiles emerge from solving this coupled system simultaneously.

Learning Objectives

After completing this example, students should be able to:

- Write mole balances for a PBR
- Account for pressure drop in a PBR
- Develop reversible rate laws
- Apply the Arrhenius equation
- Apply the van't Hoff equation
- Develop energy balances for:
    - Adiabatic operation
    - Constant-T_a heat exchange
    - Co-current heat exchange
    - Counter-current heat exchange
- Calculate equilibrium conversion
- Solve coupled ODE systems numerically

- Interpret the impact of heat transfer on reactor performance

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓

►One Reaction: Non-Isothermal Reactor Design◄

↓

Multiple Reactions

↓

Reactor Safety

↓

Challenging the Well-Mixed Assumption
(Residence Time Distributions)

↓

Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

The elementary gas phase reaction A + B ⇋ 2C is carried out in a PBR with pressure drop. Plot X, X_e, T, T_a, and $-r_A$ as a function of catalyst weight for adiabatic, co-current hx, counter-current hx, constant T_a.

Data:

F_{A0} = 5 mol/s, $F_{B0} = 2F_{A0} = F_I$.

C_{A0} = 0.2 M.

T_0 = 325 K.

T_{a0} = 300 K.

Heat capacities of all species = 20 cal/mol/K. Heat capacity of the cooling fluid is 18 cal/mol/K.

E = 25 kcal/mol

ΔH^{rxn} = -20 kcal/mol at 298 K.

$k = 0.0002 \frac{\mathrm{dm}^6}{\mathrm{kg \cdot mol \cdot s}}$ at 300 K.

α = 0.00015 kg^{-1}.

$Ua = 320 \frac{\mathrm{cal}}{\mathrm{s \cdot m^3 \cdot K}}$

$\dot{m}_c$ = 18 mol/s

ρ_b = 1400 kg/m^3

K_C = 1000 at 305 K.

Step 1: What Reactor Are We Analyzing?

This is a: ___________________________ `Reactor`

The governing mole balance is:

$$F_{A0} \frac{dX}{dW} = -r_A'$$

CBE 3610 Handbook Eq. (23).

Step 2: What Variables Change Along the Reactor?

We want to predict:

`X(W),` ___________________________, ___________________________,

___________________________, `and -rA(W)`

Step 3: Calculate Feed Conditions

Given:

$$F_{A0} = \frac{5\ mol}{s}, F_{B0} = 2F_{A0}F_{I0} = 2F_{A0}$$

Calculate the Feed Ratio: $\Theta_B = \frac{F_{B0}}{F_{A0}} = 2$ and $\Theta_I = \frac{F_{I0}}{F_{A0}} = 2$

Step 4: Write the Concentration Equations

Using:

- Pressure drop
- Variable temperature
- Stoichiometry

Use Eq. (55) in the CBE 3610 Handbook

For the reaction $A + B \rightleftharpoons 2C$ calculate:

$$\delta = \frac{2}{1} - \frac{1}{1} - \frac{1}{1} = 0$$

and therefore: $\epsilon = y_{A0}\delta = 0$

Write CA(X,T,p), CB(X,T,p), and CC(X,T,p) for this example.

CA = ___________________________

CB = ___________________________

CC = ___________________________

Step 5: Temperature-Dependent Quantities

Arrhenius Equation

Use Handbook Eq. (49):

$$k(T) = k_{ref} \exp\left[\frac{E}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right]$$

Write the expression used for this example:

k(T) = ___________________________

van't Hoff Equation

Use:

$$K_C(T) = K_{C,\text{ref}} \exp\left[\frac{\Delta H_{rxn}}{R}\left(\frac{1}{T_{\text{ref}}} - \frac{1}{T}\right)\right]$$

For this example, KC(T) = ____________________________

Step 6: Write the Rate Law

The reaction is: $A + B \rightleftharpoons 2C$

For an **elementary reversible reaction**, the rate law is:

$$-r_A = k\left(C_A C_B - \frac{C_C^2}{K_C}\right)$$

Substituting the concentration equations and temperature-dependent quantities gives:

$-r_A =$ ____________________________ where: $k = k(T)$ and $K_C = K_C(T)$

Step 7: Mole Balance

Using Handbook Eq. (23):

$$F_{A0}\frac{dX}{dW} = -r_A'$$

Step 8: Pressure Drop

Use the simplified Ergun equation.

Handbook Eq. (58):

$$\frac{dp}{dW} = -\frac{\alpha}{2p}(1 + \epsilon X)\frac{T}{T_0}$$

$\epsilon = 0$ thus, the pressure-drop equation simplifies to:

$$\frac{dp}{dW} = -\frac{\alpha}{2p}\frac{T}{T_0}$$

Substituting the numerical value of α,

$\frac{dp}{dW} =$ ______________________________

Step 9: Energy Balance (Constant T_a Case)

Handbook Eq. (33):

$$\frac{dT}{dW} = \frac{r_A' \Delta H_{rxn} - \frac{U_a}{\rho_b}(T - T_a)}{F_{A0}\left(\sum_i \Theta_i C_{P,i} + \Delta C_P X\right)}$$

For this example:

- All species have the same heat capacity.
- $\Delta C_P = 0$.
- $F_{B0} = 2F_{A0}$.
- $F_I = F_{B0}$.

Therefore:

$$\sum_i \Theta_i C_{P,i} = (1 + 2 + 2)C_P = 5C_P$$

and the energy balance simplifies to:

$$\frac{dT}{dW} = \frac{r_A' \Delta H_{rxn} + \frac{U_a}{\rho_b}(T_a - T)}{(F_{A0} + F_{B0} + F_I)C_P}$$

Where $T_a = 300$ K is constant.

Step 10: Adiabatic Case

For the adiabatic reactor: `Heat transfer term =` ________________

Therefore, the energy balance simplifies to ________________

Step 11: Co-Current Heat Exchanger

Additional Unknown: Coolant Temperature, Ta

In the co-current case, both the reactor fluid and the cooling fluid flow in the same direction, so T_a changes along the reactor and must be solved simultaneously with X, p, and T.

Handbook Eq. (34)

$$\frac{dT_\mathrm{a}}{dV} = \frac{U_\mathrm{a}(T - T_\mathrm{a})}{\dot{m}_\mathrm{C} C_{PC}}$$

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or the packed-bed reactor, convert from a volume basis to a catalyst-weight basis using the bulk density. The resulting equation is:

$$\frac{dT_\mathrm{a}}{dW} = \frac{(U_\mathrm{a}/\rho_\mathrm{b})(T - T_\mathrm{a})}{\dot{m}_\mathrm{C} C_\mathrm{PC}}$$

Boundary Condition

At the reactor inlet:

$T_\mathrm{a}(W = 0) =$ ________________

Therefore, the co-current heat-exchanger model contains four coupled ODEs:

dX/dW, dp/dW, dT/dW, dTa/dW

which must be solved simultaneously.

Step 12: Counter-Current Heat Exchanger

For a counter-current heat exchanger, the coolant flows in the opposite direction of the reacting fluid.

Handbook Eq. (35):

$$\frac{dT_a}{dV} = \frac{U_a(T_a - T)}{\dot{m}_C C_{PC}}$$

Converting to a catalyst-weight basis:

$$\frac{dT_a}{dW} = \frac{(U_a/\rho_b)(T_a - T)}{\dot{m}_C C_{PC}}$$

Boundary Condition

Unlike the co-current case, the coolant temperature is specified at the **reactor exit**:

$$T_a(W = W_f) = 300 \text{ K}$$

This means that the counter-current problem is a **boundary value problem** rather than a standard initial value problem. In practice, we often guess the inlet coolant temperature and adjust it until the desired outlet coolant temperature is obtained.

Additional Question

Why is the counter-current problem more difficult?

The coolant boundary condition is specified at the reactor exit rather than at the inlet. Therefore, the inlet coolant temperature must be guessed and adjusted until the correct outlet coolant temperature is obtained.

Step 13: Equilibrium Conversion

At equilibrium:

$$-r_A = 0$$

Therefore:

$$C_A C_B = \frac{C_C^2}{K_C}$$

Solve in terms of Xe using the quadratic formula.

Summary of ODE Systems

Constant T_a

Unknowns: `X, p, T.` The ODE system contains:

$\frac{dX}{dW}, \frac{dp}{dW}, \frac{dT}{dW}$ with a constant coolant temperature: $T_a = 300$ K throughout the reactor.

Adiabatic

Unknowns: `X, p, T.` The ODE system contains:

$\frac{dX}{dW}, \frac{dp}{dW}$, and T(X)

Co-Current Heat Exchanger

Unknowns: `X, p, T, Ta.` The ODE system contains:

$\frac{dX}{dW}, \frac{dp}{dW}, \frac{dT}{dW}$, and $\frac{dT_a}{dW}$ with initial condition $T_a(W = 0) = 300$ K

Counter-Current Heat Exchanger

Unknowns: X, p, T, Ta. The ODE system contains:

$\frac{dX}{dW}, \frac{dp}{dW}, \frac{dT}{dW}, \frac{dT_a}{dW}$ with boundary condition: $T_a(W = W_f) = 300$ K

Unlike the co-current case, the coolant temperature is specified at the **reactor exit** rather than at the reactor inlet. Therefore, the coolant inlet temperature must be guessed and adjusted until the desired outlet coolant temperature is obtained. This is why the counter-current heat exchanger is treated as a boundary-value problem rather than an initial-value problem.

Final Results

Complete after running the notebook.

Case	Final Conversion
Constant T_a	___________
Adiabatic	___________
Co-current HX	___________
Counter-current HX	___________

Reflection Questions

1. Which reactor configuration achieved the highest conversion?

Answer: Constant-Ta Heat Exchanger

The constant-T_a case achieved the highest conversion because it removed heat most effectively, keeping the reactor temperature below the equilibrium-limiting region.

Which reactor configuration achieved the lowest conversion?

Answer: Adiabatic Reactor

The adiabatic reactor achieved the lowest conversion because the temperature increased significantly, causing the reactor to reach equilibrium earlier.

Why does the reaction rate approach zero?

Answer: As temperature increases, the equilibrium conversion decreases. Eventually X approaches Xe. The driving force for reaction decreases and the net reaction rate approaches zero.

Why do the heat-exchanger cases outperform the adiabatic case?

Answer: The heat exchangers remove heat from the reactor. This keeps the temperature below
the equilibrium-limited region. The reaction remains farther from equilibrium, allowing conversion to continue increasing throughout more of the reactor.

As a result, the heat-exchanger cases achieve higher conversions than the adiabatic reactor.

In this example, the adiabatic reactor heats up rapidly, causing the equilibrium conversion to decrease and the reaction rate to approach zero. The heat exchangers help maintain a lower reactor temperature, allowing the reaction to stay farther from equilibrium and achieve greater conversion.

Engineering Insight

Higher Temperature
↓
Faster Kinetics

but

Higher Temperature

↓
Lower Equilibrium Conversion

The best reactor design balances **kinetics** and **equilibrium limitations**. Heat exchangers can improve performance by controlling temperature, preventing the reactor from reaching equilibrium too early, and allowing conversion to continue increasing throughout the reactor.